

Multiple Choice (10 marks)

1. What is the output of `4*1**4` in python?
 - (a) 4
 - (b) 1
 - (c) 256
 - (d) 16
2. Which of the following has the greatest potential for compromising a user's personal privacy?
 - (a) A group of cookies stored by the user's Web browser
 - (b) The Internet Protocol (IP) address of the user's computer
 - (c) The user's e-mail address
 - (d) The user's public key used for encryption
3. In Python 3, let `r = lambda q: q * 2`. What is `r(3)`?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
4. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.
5. A large Java program was tested extensively, and no errors were found. What can be concluded?
 - (a) All of the preconditions in the program are correct.
 - (b) All of the postconditions in the program are correct.
 - (c) The program may have bugs.
 - (d) Every method in the program may safely be used in other programs.

True / False (10 marks)

Answer True or False

6. True or False: The function $O(\log \log N)$ grows slower than $O(\log N)$ and $O(N)$.
7. True or False: In Python 3, the `isupper()` function checks if all characters in a string are uppercase.
8. True or False: Python variable names are not case-sensitive, so 'Variable' and 'variable' refer to the same variable.
9. True or False: In Python 3, performing the left shift operation $x \ll 3$ on $x = 1$ results in the value 16.
10. True or False: In Python 3, the output of `print(list[1:3])` with the given list is ['abcd', 786, 2.23, 'john', 70.2].

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to group a sequence of key-value pairs into a dictionary of lists using collections module.
12. Write a function to get an item of a tuple.

End of Paper